

General

Louvers are required to be designed for proper natural air flow inside a metal building. Mammut has two types of standard louvers fixed and adjustable. Louvers blades in a fixed louver cannot be adjusted, where as in adjustable louvers the blades can be adjusted manually, electrically or some pneumatic arrangement.

Louvers are generally used in industrial and storage buildings. They form a part of the ventilation system that is often required to reduce buildup of dust, vapor and heat inside buildings to practical levels. The use of louvers, suitably positioned in the perimeter walls of a building, in combination with roof ventilators provides continuous circulation of natural air and creates a healthier atmosphere for the building occupants.

The total effective area of louvers is equal to the free inlet area of the building minus the area of permanent opening.

The free inlet area of a building is equal to 1.50 x ventilation area.

The following equation is used to determine the effective inlet area of one louver.

$$A_{EFF} = N \times C \times L$$

Where,

N = Number of openings in louver

C = Opening depth between blades (m)

L = Opening length (m)

Note that when using an insect screen, the effective inlet area (A_{EFF}) of the louver should be further reduced by 50%.



General

Using the same building example in **section 13.7**, page 4 of 13.

A Clear Span building that is used as a factory requires ventilation using gravity ridge ventilators.

Ventilation area	Ventilation area = Throat Width x Ventilator Length x No. of Ventilators	$= 0.6 \times 3 \times 42$ $= 75.6 \text{ m}^2$
Required free inlet area	Required free inlet area = 1.5 x Ventilation Area	$= 1.5 \times 75.6$ $= 113.4 \text{ m}^2$
Effective area of louvers	Effective area of louvers = Free inlet area – area of permanent openings	$= 113.4 \text{ m}^2 - 2 \times (4 \times 5) \text{ m}^2$ $= 73.4 \text{ m}^2$
A_{EFF}	Calculating the effective inlet area of a standard Mammut fixed louver: N = 10 openings C = 0.035 m L = 1.0 m	$A_{\text{EFF}} = N \times C \times L$ $= 10 \times 0.035 \times 1.0$ $= 0.35 \text{ m}^2$
Total no. of louvers required	Total no. of louvers required	$= 73.4 / 0.35$ $= 210$
Use 210 fixed louvers (1.0 m x 1.0 m) 105 louvers on each sidewall		

Fixed louvers

Fixed louvers are made of heavy-duty natural anodized aluminum construction and are specifically designed to fit Mammut's wall panels. The frame blades and flashing of the fixed louver are made of tempered aluminum alloy extrusions meeting ASTM B 221 Alloy 6063-T6.

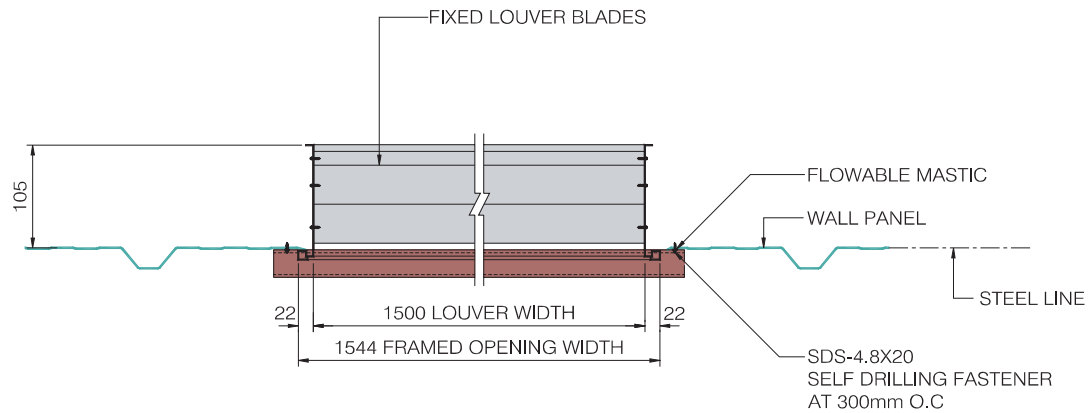
The surface of all aluminum extrusions and assemblies are anodized with a minimum anodic coating thickness of 15 microns.

The screen frames are extruded and manufactured from a suitable alloy. They are sufficiently rigid to lay flat against the louver and prevent excessive bending in frame members and sagging in the screen.

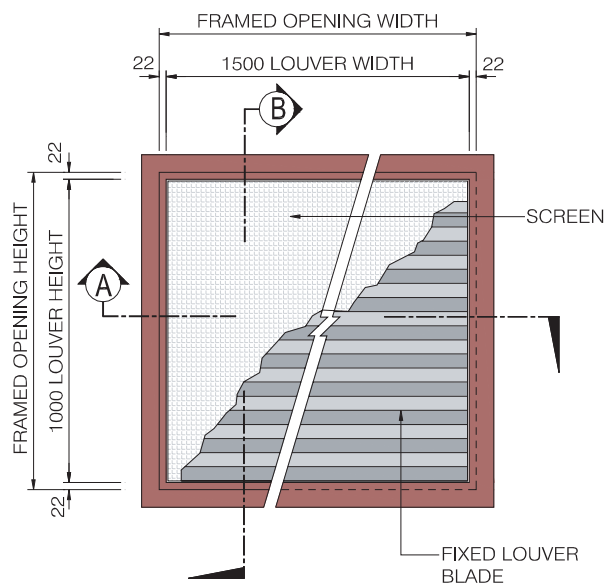
The screen and screen spine are made from aluminum or a material compatible with aluminum.

Standard fixed louvers are available in one size only: 1000 mm wide x 1000 mm high. Other custom sizes can be supplied as required.

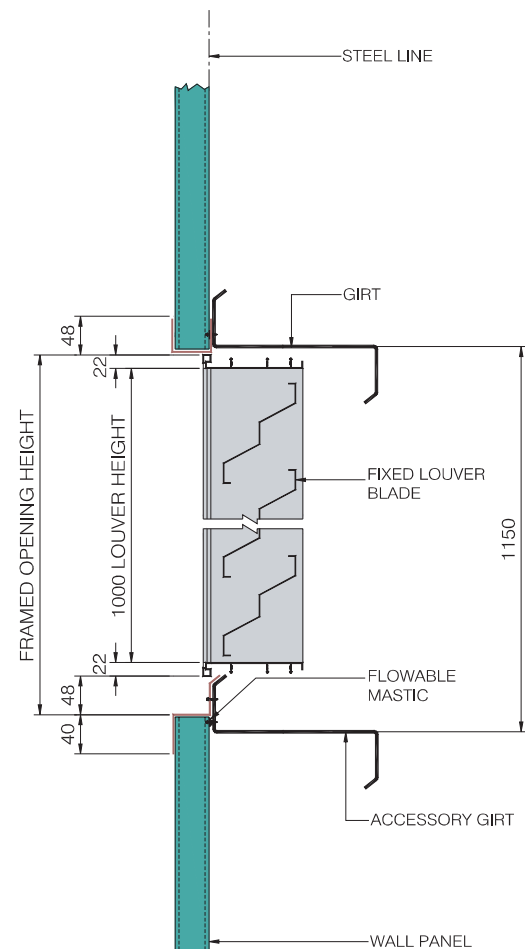




SECTION-A

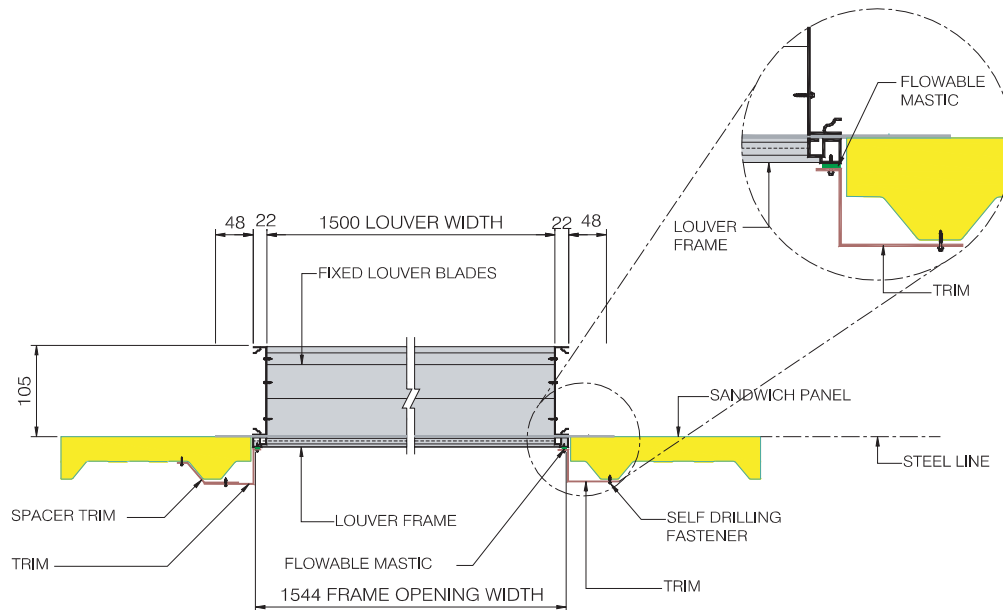


LOUVER EXTERIOR

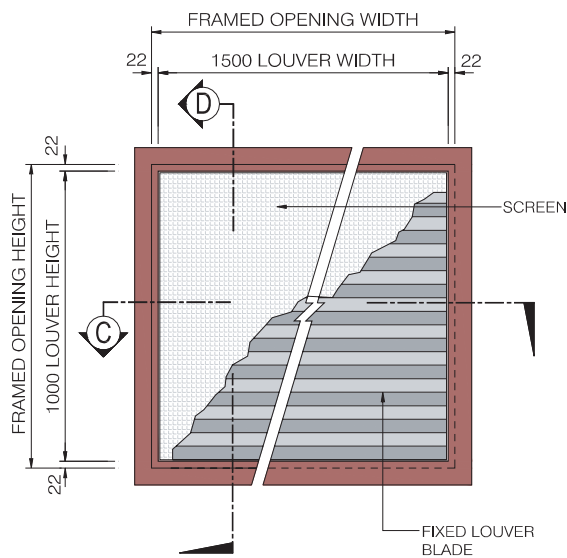


SECTION-B

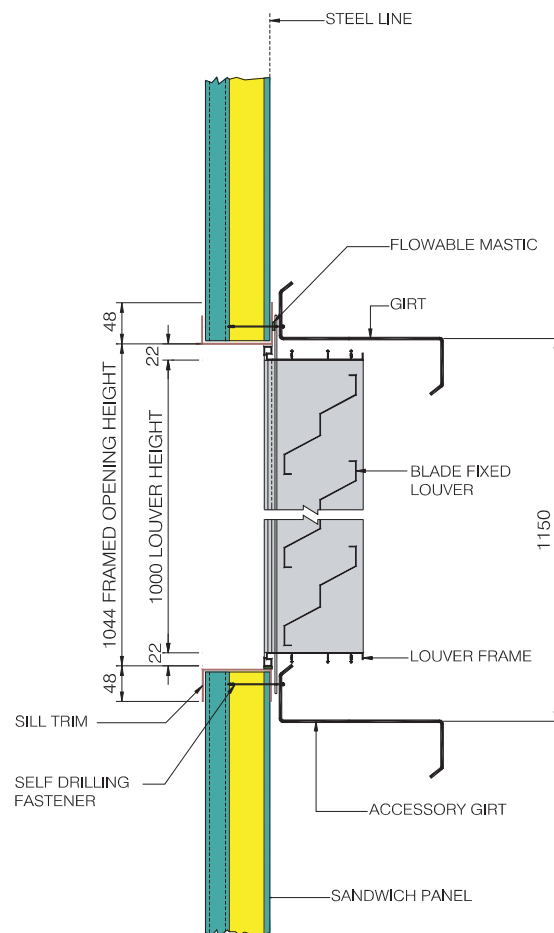
FIXED ALUMINIUM LOUVER WITH SINGLE SKIN PANEL



SECTION-C



LOUVER EXTERIOR



SECTION-B

FIXED ALUMINIUM LOUVER WITH SANDWICH PANEL

Adjustable Louvers

The adjustable louver combines fixed and adjustable blades in a single frame. Air flow control is provided while the exterior appearance remains similar to most fixed louver installations. It provides excellent weather protection, very low water penetration, high free area and low air leakage. Blade edge and jamb seals are standard. The louvers can be controlled manually, electrically or pneumatically.

Manual operation can be accomplished with a hand crank, a pull chain with spring and wall clip, or a push rod attached to the operator unit. A fusible link option can also be provided. Electrical operation utilizes motors or pneumatic which can be adapted to individual requirements.

Adjustable louvers are heavy-duty natural anodized aluminum construction and are specially designed for Mammut wall panels. Standard stationary and adjustable louvers are available for all Mammut buildings. The frame, blades and flashing of the adjustable louvers are tempered aluminum alloy extrusions confirming to ASTM B 221 Alloy 6063- T6.



Pivot clips are made of hardened aluminum to hold the blades tight.

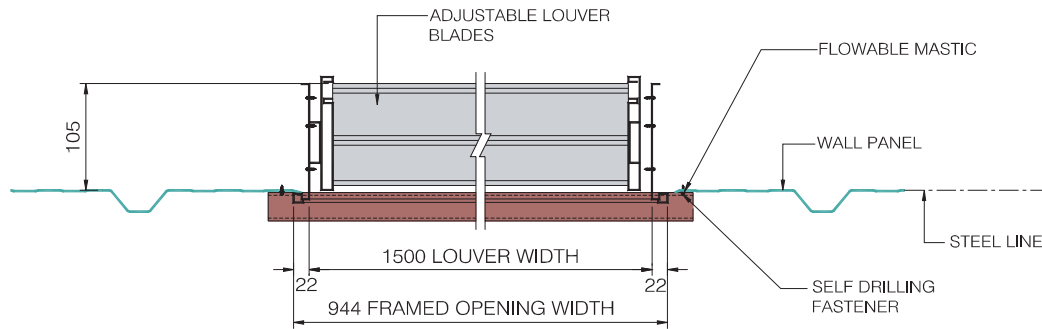
Adjustable louvers are equipped with a lever to open (full), close (tight), or set at any position in between. The louver blades operate in unison and form a tight closure. The operating mechanism should be fully lubricated during assembly. All louver blades have weather strips and, in the closed position, form a weather tight seal.

The surface of all aluminum extrusions and assemblies are anodized with a minimum anodic coating thickness of 15 microns. The screen frames are extruded and manufactured from a suitable alloy. They are sufficiently rigid to lay flat against the louver and prevent excessive bending in frame members and sagging in screening.

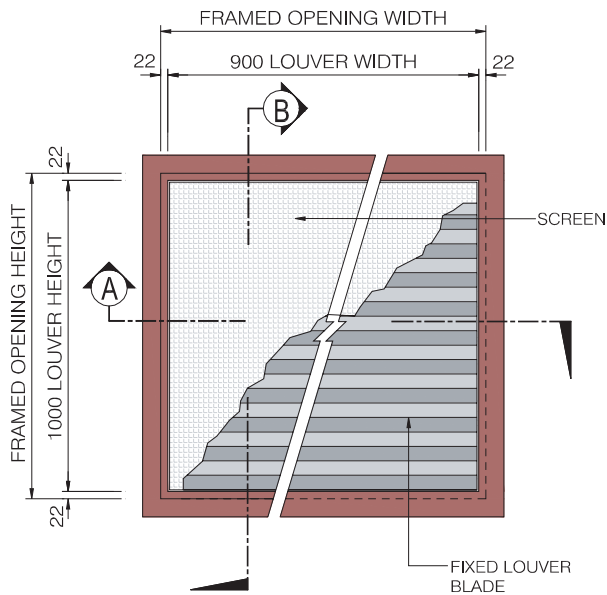
The screen and screen spine are from aluminum or a material compatible with aluminum.

Standard adjustable louvers are available in one size: 900 mm wide x 1000 mm high

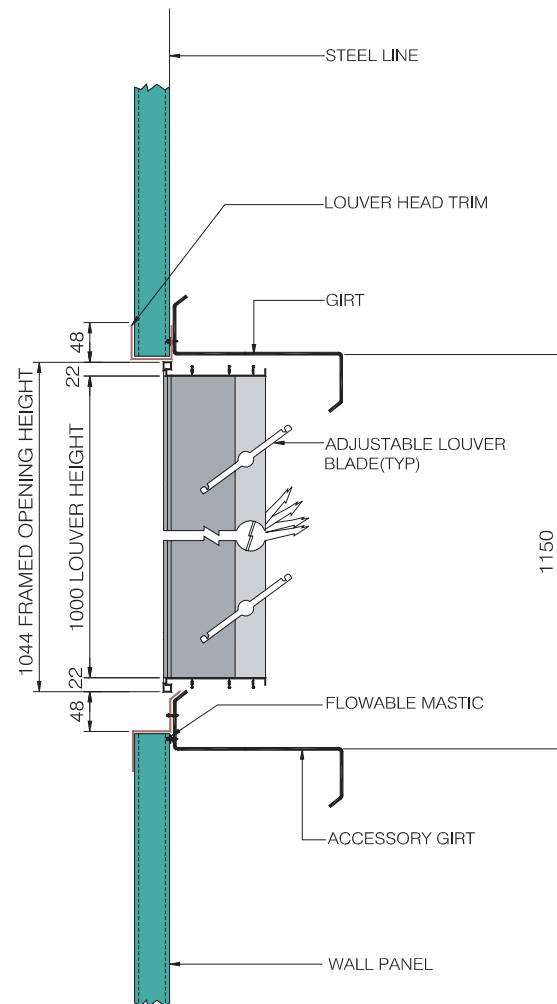
Specification	Standards
Specification for Aluminum and Aluminum-Alloy Sheet and Plate.	ASTM B209
Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes.	ASTM B221
Air Movement and Control Association, Test Method for Louvers, Dampers, and Shutters.	AMCA Standard 500



SECTION-A

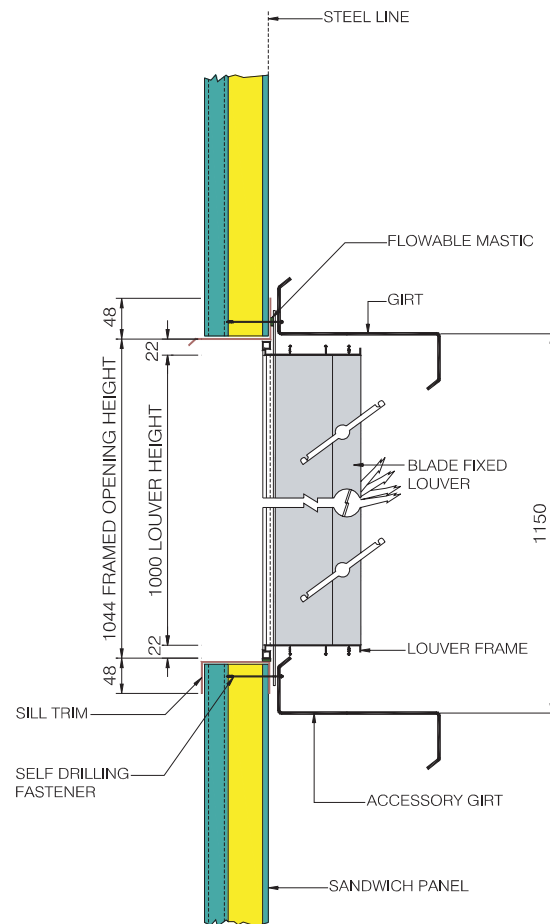
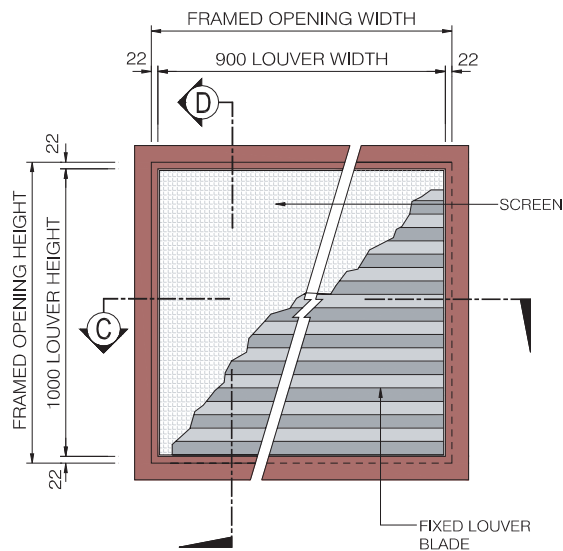
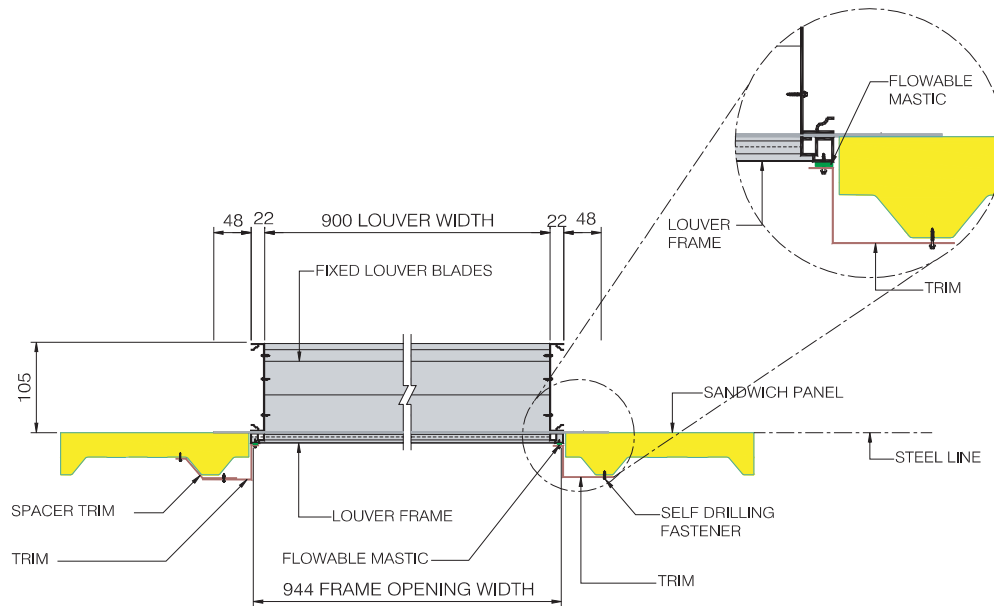


LOUVER EXTERIOR



SECTION-B

ADJUSTABLE ALUMINIUM LOUVER WITH SINGLE SKIN PANEL



ADJUSTABLE ALUMINIUM LOUVER WITH SANDWICH PANEL